$$x+1 = A(2x-3)+B$$

Comparing the coefficients of like terms, we get

$$2A = 1 \Rightarrow A = \frac{1}{2}; -3A + B = 1 \Rightarrow B = \frac{5}{2}$$

$$I = \int \frac{\frac{1}{2}(2x-3)+\frac{5}{2}}{x^2-3x+1}dx$$

$$I = \frac{1}{2}\int \frac{2x-3}{x^2-3x+1}dx + \frac{5}{2}\int \frac{1}{x^2-3x+1}dx$$

$$= \frac{1}{2}\log\left|x^2-3x+1\right| + \frac{5}{2}\int \frac{1}{\left(x-\frac{3}{2}\right)^2-\left(\frac{\sqrt{5}}{2}\right)^2}dx$$

$$= \frac{1}{2}\log\left|x^2-3x+1\right| + \frac{5}{2}\frac{1}{2\left(\frac{\sqrt{5}}{2}\right)}\log\left|\frac{x-\frac{3}{2}-\frac{\sqrt{5}}{2}}{x-\frac{3}{2}+\frac{\sqrt{5}}{2}}\right| + c$$

$$= \frac{1}{2}\log\left|x^2-3x+1\right| + \frac{\sqrt{5}}{2}\log\left|\frac{2x-3-\sqrt{5}}{2x-3+\sqrt{5}}\right| + c$$

(iii) Let $I = \int \frac{2x+3}{\sqrt{x^2+x+1}}dx$

$$2x+3 = A\frac{d}{dx}(x^2+x+1)+B$$

$$2x+3 = A(2x+1)+B$$

Comparing the coefficients of like terms, we get

$$2A = 2 \Rightarrow A = 1; \quad A+B = 3 \Rightarrow B = 2$$

$$I = \int \frac{(2x+1)+2}{\sqrt{x^2+x+1}}dx$$

$$I = \int \frac{2x+1}{\sqrt{x^2+x+1}}dx + 2\int \frac{1}{\sqrt{x^2+x+1}}dx$$

$$= 2\sqrt{x^2+x+1} + 2\int \frac{1}{\sqrt{\left(x+\frac{1}{2}\right)^2+\left(\frac{\sqrt{3}}{2}\right)^2}}dx$$

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